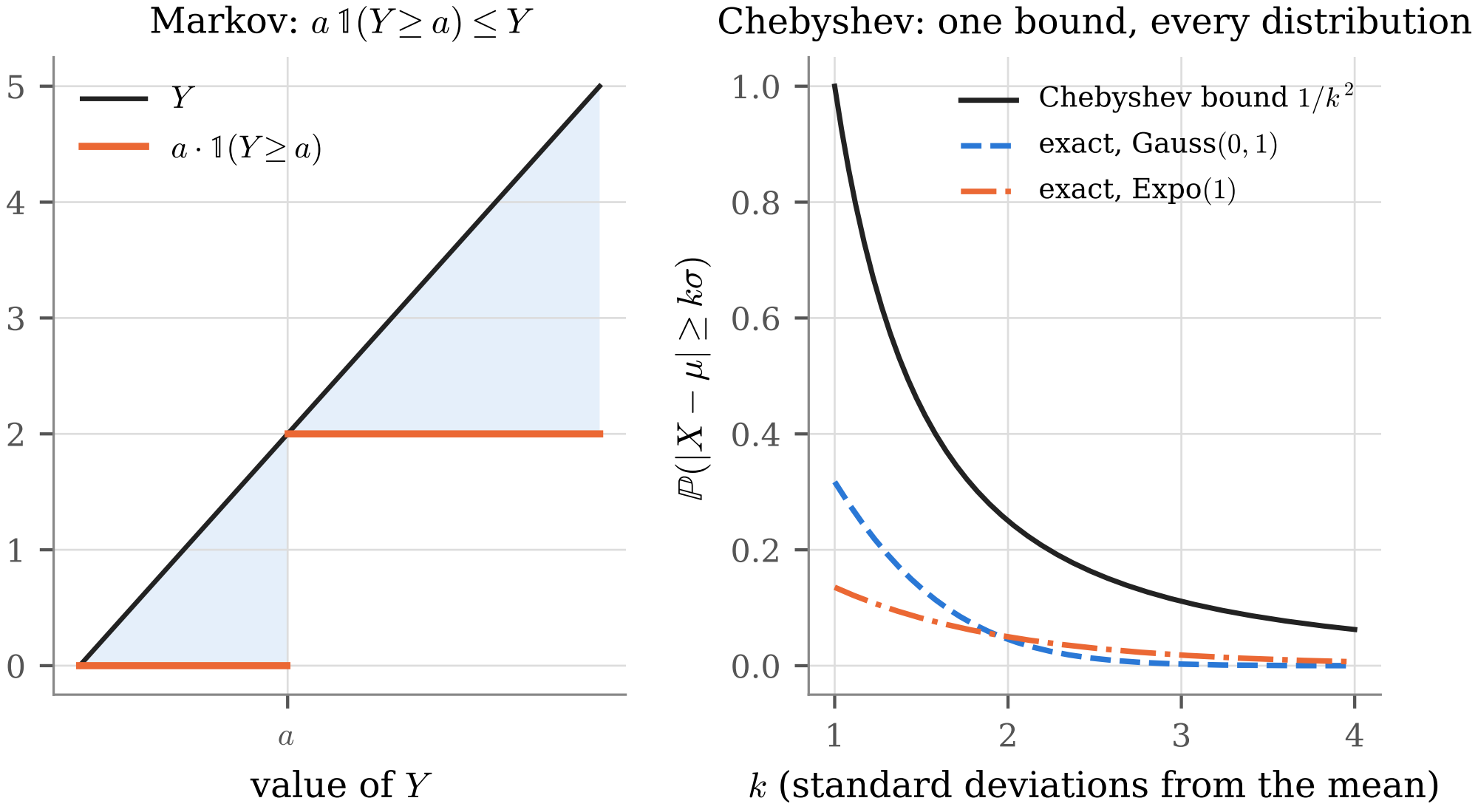
Markov: $a\,\mathbb{1}(Y \geq a) \leq Y$
Y
$a \cdot \mathbb{1}(Y \geq a)$
0
1
2
3
4
5
a
value of Y
Chebyshev: one bound, every distribution
Chebyshev bound $1/k^2$
exact, Gauss$(0, 1)$
exact, Expo(1)
$\mathbb{P}(|X - \mu| \geq k\sigma)$
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
k (standard deviations from the mean)